

Relative Neighborhood Graphs and Their Relatives

JERZY W. JAROMCZYK AND GODFRIED T. TOUSSAINT

Invited Paper

This is a survey of results on neighborhood graphs. The paper discusses properties, bounds on the size, algorithms, and variants of the neighborhood graphs. Numerous applications including computational morphology, spatial analysis, pattern classification, and data bases for computer vision are described. A rich bibliography of the subject concludes the paper.

Keywords—Computational geometry, computational morphology, geometric graphs, neighborhood graphs, spatial analysis.

I. INTRODUCTION

A decade has passed since the paper "The Relative Neighborhood Graph of a Finite Planar Set" [1] appeared in print. This anniversary provides a good opportunity to review the results obtained so far and the current state of research on neighborhood graphs.

In fact, during this time the original notion of neighborhood has been generalized in several directions and all of these graphs are now jointly referred to as proximity graphs [2]. So much interest has been spawned in this area that in December 1989 an entire conference on proximity graphs took place in Las Cruces, New Mexico [3].

The relative neighborhood graph of a finite set of points V , $RNG(V)$, is a prominent representative of the family of graphs which are defined using some concept of neighborliness. For points in a real space R^d the relative neighborhood graph of V is a graph with vertex set V and set of edges $RNG(V)$ which are exactly those pairs (p, q) of points for which $\delta(p, q) \leq \max_{v \in V \setminus \{p, q\}} \{\delta(p, v), \delta(q, v)\}$, where δ denotes the distance between p and q . Fig. 1 illustrates a set of points and their relative neighborhood graph.

The main objective of this paper is to summarize efforts of the last ten years of research on concepts which emerged while studying relative neighborhood graphs.

From a mathematical and algorithmic point of view, neighborhood graphs fall under the scope of *computational*

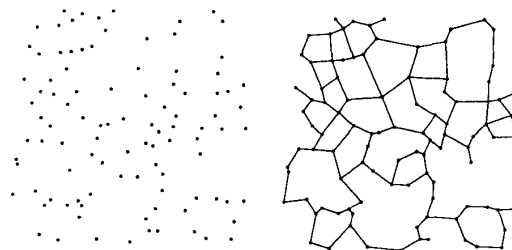


Fig. 1. A point set and its RNG.

geometry. More particularly, since they are concerned with extracting the shape and structure of points sets, from an application perspective, they form an indispensable tool of *computational morphology*. Neighborhood graphs serve as tools in disciplines where shape and structure are vital. These include visual perception, computer vision and pattern recognition, geography and cartography, and biology, to list a few; for further examples refer to the section on applications and the bibliography, which includes numerous references.

There is, however, much beyond purely practical applications which makes neighborhood graphs attractive objects of study. We will demonstrate both practical and theoretical aspects in further sections, where we discuss algorithms, properties, and applications.

II. DEFINITIONS

We will start with a definition of neighborhood graph in a quite general form. This will later enable us to introduce specialized versions more uniformly. Although our later discussion mainly pertains to R^2 and R^3 the definition is valid in any dimension.

Let V be a set of points in R^d . Each (unordered) pair of points $(p, q) \in V \times V$ is associated with a neighborhood $U_{p,q} \subset R^d$. Let $\mathcal{P}$ be a property defined on $\mathcal{U} = \{U_{p,q} : (p, q) \in V \times V\}$. A neighborhood graph $G_{\mathcal{U}, \mathcal{P}}(V, E)$ defined by the property $\mathcal{P}$ is a graph with the set of vertices

Manuscript received January 21, 1990; revised April 16, 1992.
J. W. Jaromczyk is with the Department of Computer Science, University of Kentucky, Lexington, KY 40506-0027.
G. T. Toussaint is with the School of Computer Science, McGill University, 3480 University Street, Montreal, Quebec, Canada H3A 2A7.
IEEE Log Number 9204424.

V and the set of edges E such that $(p, q) \in E$ if and only if $U_{p,q}$ has property P . If (p, q) is an edge, denoted later simply by pq , then we say that q is a neighbor of p (and vice versa). For some neighborhood graphs it is more natural to associate neighborhoods with points rather than with pairs of points.

Technically, it is convenient to differentiate between the graph and its geometric realization, which is called a neighborhood skeleton. The neighborhood skeleton of V is obtained by connecting, with straight line segments, the pairs of points which form edges in the corresponding neighborhood graph. We will use both terms alternatively without, we hope, any confusion. It is worth noting that the term *neighborhood graph* is also used, with a different and unrelated meaning, in graph theory; see [4].

The neighborhood of an edge is usually defined using the concept of distance. In this paper, we will use the metrics L_p , $1 < p < \infty$, L_1 , and L_∞ to measure the distance $\delta(x, y)$ between points $x = (x_1, \dots, x_d)$ and $y = (y_1, \dots, y_d)$ in R^d . The distance in the metric L_p is defined as $\delta_p(x, y) = \sum_{i=1}^d |x_i - y_i|^{1/p}$. In L_1 and L_∞ the distance is defined by $\delta_1(x, y) = \sum_{i=1}^d |x_i - y_i|$, and $\delta_\infty(x, y) = \max_{1 \leq i \leq d} |x_i - y_i|$ respectively. Some concepts and results presented in the paper hold for more general metrics. The distance $\delta(p, q)$ will be also called the length of pq . Furthermore, $B(x, r)$ denotes an open sphere centered at x with radius r , i.e., $B(x, r) = \{y : \delta(x, y) < r\}$. A closed sphere is defined as $\bar{B}(x, r) = \{y : \delta(x, y) \leq r\}$. In R^2 both a sphere and its boundary will be called, without any confusion, a circle.

Below we will define, within this framework, relative neighborhood graphs, Gabriel graphs, and β skeletons. Many interesting variants will be given in Section VI. We begin with an alternative definition of the *RNG*.

A. Relative Neighborhood Graphs

Let $\Lambda_{p,q} = B(p, \delta(p, q)) \cap B(q, \delta(p, q))$; $\Lambda_{p,q}$ is called a lune. The relative neighborhood graph of V , the *RNG*(V), is a neighborhood graph with the set of edges defined as follows:

$$(p, q) \in E \text{ if and only if } \Lambda_{p,q} \cap V = \emptyset. \quad (1)$$

Note that the above definition is equivalent to one given in the introduction.

As a historical digression, let us mention that lunes, as intersections of circles, have been the object of intense study since the times of ancient Greeks. In contrast to a circle (which cannot be squared with a straight edge and compass) Hippocrates of Chios (circa 440 B.C.) showed that certain types of lunes could be squared [5]. Furthermore, the type of lune used in the *RNG*, traditionally referred to as the *Vesica Piscis*, was used with symbolism as a basic design element in the floor plans of gothic cathedrals [6].

It is immediate that an edge pq is in the *RNG*(V) if there is no triangle Δpqv , $v \in V \setminus \{p, q\}$, with pq the strictly longest edge. In Euclidean spaces, relative neighborhood graphs can be equivalently defined by means of angles; we will return to this issue in Section V.

Note that if we change slightly the definition of the neighborhood by using the intersection of closed spheres (rather than their interiors) we obtain a different class of graphs. In fact such a class was defined by Lankford [7] in 1969 and historically relative neighborhood graphs could be viewed as a modification of this class (see [1]). As we will point out in later sections this modification leads to different geometric properties.

B. Gabriel Graphs

The neighborhood, called a diameter sphere, is defined as a sphere: $\Gamma_{p,q} = B(\frac{p+q}{2}, \frac{\delta(p,q)}{2})$. The Gabriel graph of V , $GG(V)$, is a neighborhood graph with the set of edges

$$(p, q) \in E \text{ if and only if } \Gamma_{p,q} \cap V = \emptyset. \quad (2)$$

In the Euclidean space, pq is an edge in $GG(V)$ if there is no triangle Δpqv , $v \in V \setminus \{p, q\}$, with $\angle pvq > \frac{\pi}{2}$. Yet another equivalent definition is that

$$pq \in GG(V) \text{ if } \delta_2(p, q) \leq \min \left\{ \sqrt{\delta_2^2(p, s) + \delta_2^2(s, q)} : s \in V \right\}. \quad (3)$$

Gabriel graphs were introduced by Gabriel and Sokal [8] in a context of geographic variation analysis.

C. β Skeletons

Kirkpatrick and Radke [9] defined a parameterized family of neighborhood graphs called β skeletons. The neighborhood $U_{p,q}(\beta)$ is defined, for any fixed $\beta (1 \leq \beta < \infty)$, as the intersection of two spheres:

$$U_{p,q}(\beta) = B\left(\left(1 - \frac{\beta}{2}\right)p + \frac{\beta}{2}q, \frac{\beta}{2}\delta(p, q)\right) \cap B\left(\left(1 - \frac{\beta}{2}\right)q + \frac{\beta}{2}p, \frac{\beta}{2}\delta(p, q)\right). \quad (4)$$

The (lune-based) β skeleton of V , $G_\beta(V)$, is a neighborhood graph with the set of edges defined as follows:

$$(p, q) \in E \text{ if and only if } U_{p,q}(\beta) \cap V = \emptyset. \quad (5)$$

A useful feature of this parametrized family is its monotonicity with respect to β , i.e. $G_{\beta_1}(V) \subset G_{\beta_2}(V)$ for $\beta_1 > \beta_2$. It is easy to see that β skeletons contain both relative neighborhood and Gabriel graphs as special cases. Specifically, the *RNG*(V) = $G_2(V)$ and $GG(V)$ = $G_1(V)$. In fact, as we will see in Section V, it is possible to design a uniform algorithm for the whole spectrum of β skeletons for $1 \leq \beta \leq 2$.

The β skeletons have interesting applications to the analysis of interpoint linkages in empirical networks. Kirkpatrick and Radke [9] illustrated such an analysis, showing examples of road and airline networks. In particular, a comparison of links in an empirical network with the set of edges in the computed β skeletons for various values of β helps to detect significant patterns.

Lune-based β skeletons can be also defined for $0 < \beta < 1$. To this end, the neighborhood of (p, q) is defined as the intersection of two spheres of radius $\delta(p, q)/2\beta$ which

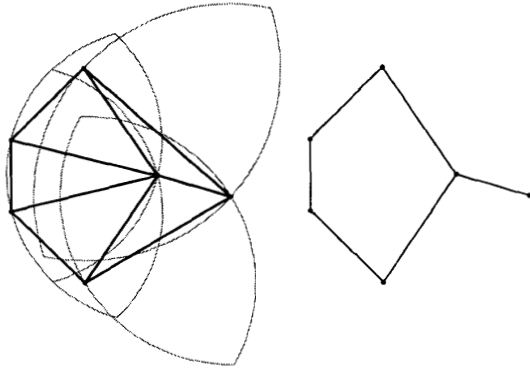


Fig. 2. $DT(V)$ with four lunes and the $RNG(V)$.

contain p and q in their boundaries. In fact this extension leads to so-called circle-based β skeletons. For $\beta \geq 1$, the neighborhood $\mathcal{U}_{p,q}$ of (p, q) is defined as the union of two spheres of radius $\beta \times \delta(p, q)/2$ passing through p, q . The points p, q are connected with an edge if the neighborhood is empty. The circle-based β skeleton for $\beta \geq 1$ can be constructed in $O(n \log n)$ time; see [9].

III. PROPERTIES

Relative neighborhood graphs are related to other prominent geometric structures such as minimum spanning trees (MST) and Delaunay triangulations (DT). The Delaunay triangulation of a set V is defined as the dual graph of the Voronoi diagram of V which is a decomposition of R^d into n cells, n being the number of points in V . A point x is in the cell associated with $v \in V$ if for all $w \in V \setminus \{v\}$, $\delta(x, v) < \delta(x, w)$. Two points in the $DT(V)$ are connected with an edge if the boundaries of their Voronoi cells intersect. For definitions and properties of the MST and the DT see for example [10]–[12].

Toussaint showed that in the Euclidean plane $MST(V) \subset RNG(V) \subset DT(V)$. This implies, in particular, that the RNG is connected. Fig. 2 illustrates the Delaunay triangulation of a point set and nonempty lunes for those of its edges which are not in the RNG .

These inclusions can be generalized to other metrics and higher dimensions. A crucial step in this direction was made by O'Rourke [13]. Studying the RNG in the L_1 metric, he noticed that the dual graph of the Voronoi diagram is not necessarily a supergraph of the RNG in L_1 and L_∞ . Instead, he proposed a definition of the DT directly in terms of the points V . Two points in V are connected by an edge in the DT if there exists a sphere (with respect to the metric at hand) such that its boundary contains these points and no point of V is in the interior of this sphere. This definition is equivalent to the traditional one for the metric L_p in R^2 space (see [13] and [14]). Here, for the sake of uniqueness of the DT , a general position of points is assumed. In our context we say that points are in a general position in R^d if no $d+1$ of them are coplanar (lie on a common $d-1$ flat) and no $d+2$ of them are cospherical (cocircular) with

respect to the given metric. In some cases it will be possible to relax the general position assumption.

This definition of the DT allows extension of the result of Toussaint [1] that the RNG of a subgraph of the DT to all of the metrics L_p , L_1 , and L_∞ and to an arbitrary dimension; see [13].

Similar relationships hold for β graphs in L_p . More specifically, for R^2 and L_p we have $MST(V) \subset G_\beta(V) \subset DT(V)$ ($1 \leq \beta \leq 2$) (see [15] for L_p and [9] for L_2). In particular, the relation holds for Gabriel graphs; see also Matula and Sokal [16].

As we will see in further sections, the fact that the DT is a supergraph of the RNG (and β graphs) turns out to be very useful in designing efficient algorithms for neighborhood graphs.

IV. SIZE OF NEIGHBORHOOD GRAPHS

It is both interesting and important to know lower and upper bounds for the number of edges in the $RNG(V)$. The results of the previous section give immediate bounds in two-dimensional space. Since the MST is a subgraph of the RNG and the RNG is a subgraph of the DT , we have immediately that $|MST(V)| \leq |RNG(V)| \leq |DT(V)|$. Expressing this in terms of $n = |V|$ we have in L_p that $n-1 \leq |RNG(V)| \leq 3n-6$. The upper bound follows from the planarity of the $DT(V)$. Note that a similar argument works for β graphs as well. A more detailed analysis gives the tighter bounds of $3n-8$ for the GG (see [16]) and $3n-10$, $n \leq 8$, for the RNG (see Urquhart [17]) in the Euclidean plane. The bounds for the RNG and the GG are tight for an infinite number of n . In the metrics L_1 and L_∞ the RNG can have $\Theta(n^2)$ edges; an example has been given by Katajainen [18].

Consider now the case of Euclidean R^d space. As higher dimensional spaces are easier to discuss, we will begin with $d \geq 4$. A discussion on the three-dimensional case will be deferred.

The first interesting observation is that the relative neighborhood graphs can be dense in R^d , $d > 3$. The maximum number of edges of the relative neighborhood graph of n points in R^d , $d \geq 4$, is $\Omega(n^2)$.

Consider a set V in R^4 that contains an even number of points with $n/2$ points of the form $(a, b, 0, 0)$, where $a^2 + b^2 = 1$, and $n/2$ points of the form $(0, 0, c, d)$, where $c^2 + d^2 = 1$. Each pair of points from different groups forms an edge of the $RNG(V)$; this shows that the size of the RNG can be $\Omega(n^2)$. Note that the points are not in general position. By embedding the set V into R^d we obtain this quadratic bound for any $d \geq 4$. Clearly, the same $\Omega(n^2)$ bound holds for all β graphs. The above construction, based on Lenz's example, is standard and has been used by many authors for various spatial graphs.

The problem of estimating the sizes of the RNG and the GG in three-dimensional space is much more interesting. Let us start with the RNG ; again we focus on Euclidean space. The following result was the first subquadratic upper bound on the size of the RNG (see [19]): the maximum

number of edges of the relative neighborhood graph of n points in $d \geq 3$ dimensions is $O(n^{3/2+\epsilon})$ for each $\epsilon > 0$.

This result can be established using upper bounds on the size of *unit distance graphs*. The unit distance graph of V is a graph with two points connected by an edge if they are in a particular, say, one unit, distance. Recent results by Clarkson *et al.* [20] show that the size of the unit distance graph of an n point set is bounded by $cn^{3/2+\epsilon}$, where $\epsilon > 0$ is an arbitrary small real number. In fact, they give this bound in a stronger form of $O(n^{3/2}(\lambda_6(n)/n)^{1/4})$, where λ_6 is related to the complexity of Davenport-Schinz sequences. For more discussion on the size of unit distance graphs in Euclidean spaces, see also Chung [21].

To derive an upper bound, decompose the *RNG* into subgraphs of edges of the same length. For each subgraph an upper bound can be established based on the results of the size of unit distance graphs. Furthermore, the number of different lengths of edges in the *RNG*(V) which are adjacent to any vertex is bounded by a constant independent of V . Hence each vertex participates in a constant number of subgraphs and an upper bound of $O(n^{3/2+\epsilon})$ for the total number of edges in the *RNG* follows. Clearly, any improvement on the upper bound on the size of unit graphs automatically improves the upper bound on the *RNG*.

As a remark in passing let us mention that subgraphs of the *RNG* of edges of the same length have other interesting properties. For example the interior of a pyramid spanned by three edges of such a subgraph that share a common endpoint has an empty intersection with the remaining edges of this subgraph. We call this a nonpenetration property.

Agarwal and Matoušek [22] proved an upper bound of $O(n^{4/3})$ using bichromatic closest neighbor pairs and an interesting technique of computing a “small” family of well-separated pairs which include all edges in the *RNG* [23]. Let P, Q be two point sets. We say that points $p \in P, q \in Q$ form a bichromatic closest neighbor pair if $\delta(p, q) = \min_{x \in P, y \in Q} \delta(x, y)$. Furthermore, we say that P and Q are well separated if there exists a pair of cones C, C' with a common apex, a common axis, and the angle at the apex less than $\pi/3$, such that $Q \subset C$ and $P \subset C'$. The following holds [22]: if P, Q are well separated in R^d , then *RNG*($P \cup Q$) has an edge $pq, p \in P, q \in Q$ if and only if (p, q) is a bichromatic closest neighbor pair. Based on the result of Edelsbrunner and Sharir [24], the number of edges in the *RNG*($P \cup Q$) in R^3 , with P and Q well separated, is $O(n^{2/3}m^{2/3} + m + n)$, $n = |P|, m = |Q|$. Next, it is possible to cover $V \subset R^3$ by a family of well-separated pairs $(P_1, Q_1), \dots, (P_s, Q_s)$ with two properties: (i) for each $p, q \in V$, there is (P_i, Q_i) with $p \in P_i, q \in Q_i$; (ii) $\sum_{i=1}^s |P_i| + |Q_i| = O(n \log^2 n)$. Each edge in the *RNG* connects a pair of bichromatic neighbors in some well-separated pair of this family. Hence counting the number of bichromatic neighbor pairs in all of the well-separated pairs gives an upper bound on the size of the *RNG*. Concluding: *the maximum number of edges of the relative neighborhood graph of n points in R^3 is $O(n^{4/3})$.*

Again, any improvement in the upper bound on the size

of bichromatic closest neighbor pairs would improve the upper bound on the *RNG*.

We do not know about any nontrivial lower bound on the size of the *RNG* in R^3 . For the special case of point sets that do not allow equal distances between pairs of points, it is easy to show that the size of the *RNG* is linear. This observation also applies to point sets for which the number of equidistant neighbors for each vertex is bounded by a common constant; in particular it holds for points in general position. A geometric fact worth noting is that the degree of each vertex of the *RNG*, if no isosceles triangles are formed, is bounded by a constant. This is also true for $L_p, 1 < p < \infty$ (see for example [25]). Therefore $|RNG(V)| \leq c|V|$ for point sets in R^d that do not allow equal distances between pairs of points. Alternatively, the size of relative neighborhood graphs defined by closed lunes is linear.

A. Size of Gabriel Graphs

As we remember, Gabriel graphs are supergraphs of relative neighborhood graphs. It appears that Gabriel graphs can have substantially more edges than the *RNG* already in R^3 . There are point sets in R^3 whose Gabriel graphs have $\Omega(n^2)$ edges (see [26] and [27]). Therefore the following results holds: *the maximum number of edges of the Gabriel graph of n points in $d \geq 3$ dimensions is $\Omega(n^2)$.*

An example in [27] gives a point set V which consists of two groups of $n/2$ points located on two circles. These circles are placed in two orthogonal planes and they pass through each other's center. The points are located close to these centers; this distance can be precisely specified. A construction which uses parabolas, and therefore avoids many cocircular points, was provided in [26]. Consider $2n$ points on two parabolas; n of them on $x = 1 - z^2/4, y = 0$ at $z = 0 \pm \epsilon, \pm 2\epsilon, \dots$ and the remaining n points on the parabola $x = y^2/4 - 1, z = 0$, at $y = 0, \pm \epsilon, \pm 2\epsilon, \dots$, where ϵ is a sufficiently small positive number. Each of the points is extremal on the convex hull of this set of points. With a small perturbation of the points we can preserve density of the *GG* while placing the points in general position.

Interestingly, for any set of n points P in $d \geq 3$ dimensions there is a set Q of $O(\sqrt{n} \log^{d-1} n)$ points so that the Gabriel graph of $P \cup Q$ has at most $O(n^{3/2} \log^{d-1} n)$ edges (see [27]). Thus by adding extra points we can reduce the size of certain Gabriel graphs substantially; furthermore this set of points can be effectively found.

B. Expected Values for Proximity Graphs

The expected size of Gabriel graphs and relative neighborhood graphs in Euclidean R^d space for various set distributions has been studied, among other graphs, by Devroye [28]. He showed that for all densities $\liminf E(|GG(V)|)/|V| \geq 2^{d-1} E(\gamma)$ being the expected value of a given random variable γ . Furthermore, for most densities $E(|GG(V)|) \sim 2^{d-1} |V|$. This result extends a similar result by Matula and Sokal [16], who demonstrated that for points uniformly distributed in a unit square $E(|GG(V)|) \sim 2|V|$.

For relative neighborhood graphs the expected size is estimated by $E(|RNG(V)|)/|V| \geq C_d + o(1)$, where C_d is a constant that depends only on d . In particular, for $d = 2$ the expected number of edges in the $RNG(V)$ is at least $(1.27 + o(1))|V|$.

Several results concerning the expected values for the RNG and the GG , as well as many other geometric graphs, in the unit density Poisson probabilistic model were given in [26].

Let us mention that considerable work has also been done on computing expected values of properties such as the expected length, perimeter, and area of a triangle in the DT . These issues are well covered in Getis and Boots [29]; see also [30].

V. ALGORITHMS

A variety of algorithms have been proposed for the RNG in the literature. They are based on ideas developed independently by several researchers. From a retrospective we can identify techniques which are common to several approaches.

A paradigm used in most of the algorithms is a refinement approach. It fits to a framework of *filtering* (for a general treatment and development of this technique and its application to many fundamental geometric problems see Chazelle [31]). Applications of filtering for the RNG have been described by Katajainen [32].

In this approach the RNG is built in stages (usually two) where a sequence of supergraphs with the RNG as a final graph is constructed. Each supergraph is obtained from the previous one by eliminating edges which cannot belong to the RNG . This elimination, called also a pruning, is either based directly on the definition or uses geometric properties of relative neighborhood graphs.

Often such a process applies other fundamental algorithms, as well as nontrivial data structures and general algorithm design techniques.

In fact, common to many algorithms for neighborhood graphs is a general method called the region approach (see [33] for various applications). The main idea of the region approach is to divide space in such a way that each point is associated with a finite number of simple regions and the search for the neighbors of the point in these regions reduces to simple queries. One of the most important general references here is Yao's paper [34], which presents subquadratic algorithms for various proximity problems in R^d , e.g., *general geographic neighbors* which are particularly useful in algorithms for neighborhood graphs. In addition, the paper illustrates another powerful technique, *balancing*, i.e., how to trade preprocessing and query costs to minimize the overall processing time.

Not surprisingly, particularly useful for constructing neighborhood graphs are *point location* and *range searching* algorithms. *Point location* consists in identifying the region, in a partition of the geometric space, where the query point is located. *Range search* is a problem of counting or retrieving all points from a given collection of points which

are contained in the query region. A typical query region can be a sphere, a lune, a simplex, a halfspace, or a box.

The point location or range-search approach is a natural implication of the way neighborhood graphs are defined; a search of the neighborhood of an edge gives complete information as to whether this edge has to be eliminated. The power of this approach for constructing the RNG was recognized by O'Rourke [13]. He employed an efficient data structure for range queries to construct the RNG in the L_∞ (for arbitrary dimension) and L_1 metrics. The point location approach was also taken by Toussaint, Bhattacharya, and Poulsen [35] to obtain a practically efficient algorithm for computing the Gabriel graph in all dimensions.

As we will see, the difficulty of the RNG problem will depend on whether the given points are in general position. Recall that in our context we say that points are in general position in R^d if no $(d+1)$ of them belong to a common $(d-1)$ flat and no $(d+2)$ of them are cospherical (cocircular) with respect to a given metric.

A. Two-Dimensional Space

A straightforward approach to constructing the RNG is to check, for each edge and point, whether this point is in the neighborhood of the edge. The cost of such a procedure is obviously $\Theta(n^3)$ but this approach does work for any metric or dimension. (In fact, it works for any neighborhoods for which a membership test costs $O(1)$.)

One can immediately reduce this complexity by a factor of n recalling that the Delaunay triangulation (which has $O(n)$ edges) is a supergraph of the relative neighborhood graph of a given point set; this observation led to the first $O(n^2)$ algorithm for the RNG [1].

Clearly the $DT(V)$ is not the only supergraph that can be effectively utilized for constructing the RNG . A family of such graphs can be obtained using a general approach of *frames* and *narrow regions* (see Yao [34] and Gabow, Bentley, and Tarjan [33]). A region associated with a point v is narrow if for $p, s \in V$ that belong to this region we have $\delta(p, s) < \max\{\delta(v, p), \delta(v, s)\}$. Such regions can be obtained, for example, by dividing the space around a given point p into equal sections by eight lines passing through p . In particular, Katajainen [18] used octant neighbors as a supergraph of the RNG . An octant (geographic) neighbor of p in a region $r(p)$ associated with p is a point of $V \setminus \{p\}$ in $r(p)$ which is closest to p . The octant neighbor graph is obtained by connecting each point of V with its geographic neighbors in all regions. Since an octant neighbor graph can be decomposed into eight planar graphs, its size is linear (in $L_p, 1 < p < \infty$). Phasing out edges from this graph by testing them against each point leads to an $O(n^2)$ algorithm.

Using octant neighbors and a range query algorithm, e.g. [31], [33] [36] for rectangles (intersections of spheres in L_1 and L_∞), Katajainen [18] derived an $O(n \log n + m)$ output sensitive algorithm for the RNG in L_1 and L_∞ , m being the output size. To achieve a linear space the range searching is organized accordingly to the batching technique described in [37].

For further results regarding geographic neighbors or the *angle restricted nearest neighbor* see also [38] and [39].

Probably the simplest way to divide the plane into regions is to use a square grid. The *cell technique* (see Bentley, Weide, and Yao [40]), which leads to a fast expected time algorithm for many closest point problems, has been used by Katajainen, Nevalainen, and Teuhola [41], [42] (see also [43]) to design a linear expected time algorithm for the *RNG* in a Euclidean plane. They show that the expected time is linear for the points (in a unit square) generated by a homogeneous Poisson process. Note that the *floor* function is essential in applying the cell technique.

More Elaborate Elimination Techniques: The first $O(n \log n)$ algorithm for the *RNG* in L_2 was developed by Supowit [44]. Supowit was able to organize the elimination of edges from the $DT(V)$ in $O(n \log n)$ time, matching the cost of building the $DT(V)$. The elimination uses the sweep line technique (see, e.g., [10] and [11]; the sweeping is done in six directions (multiples of $\frac{\pi}{3}$). Assume that we sweep from right to left in the direction parallel to a line l . A sweep status structure T maintains *active* points (at the beginning T is empty); E is set to the $DT(V)$. During the sweep any encountered point $p \in V$ is inserted into T . Then, for edges e with the left point p (in the order of increasing angles with respect to l) and for any point $v \in T$ such that the horizontal line passing through v intersects e , we check if v eliminates e . If e is eliminated then e is removed from E ; otherwise v is deleted and the plane is swept further. The above process, repeated in the six directions, is able to extract all edges of the $DT(V)$ which do not belong to the $RNG(V)$ and can be implemented to take $O(n \log n)$ time. This gives an $O(n \log n)$ algorithm, which is optimal. In fact Supowit demonstrated a simple linear time reduction of sorting to the *RNG* in one-dimensional Euclidean space; the reduction is also valid in L_p .

Using a similar approach, Lee [45] has designed an $O(n \log n)$ time algorithm to construct the *RNG* in L_1 and L_∞ metrics. The starting graph is again the Delaunay triangulation, and the assumption that no four points in V are cocircular (in the sense of L_1) is essential.

Construction of the GG in the Euclidean plane is simpler. The first optimal $O(n \log n)$ algorithm was given by Matula and Sokal [16], where applications to geographic variation research and clustering were discussed. The algorithm is based on an observation that the Gabriel graph of V contains those edges in the $DT(V)$ which do not intersect their dual Voronoi edges: see Fig. 3. Since the Voronoi diagram and $DT(V)$ can be constructed in $O(n \log n)$ time, the bound follows. Clearly, the method can be extended to R^d .

Another elimination strategy for the *RNG* has been proposed in [25] and later improved and extended by Jaromczyk, Kowaluk, and Yao [15]. This strategy is based on an interesting structure of edges in the $DT(V)$ that do not belong to the $RNG(V)$. Pick a triangle Δ in the $DT(V)$ and one of its vertices v . Then check if v eliminates the opposite edge of Δ ; call this edge e . If yes, then analyze if any two edges of the triangle adjacent to Δ along the edge e are eliminated by v . Continue this process until no edge

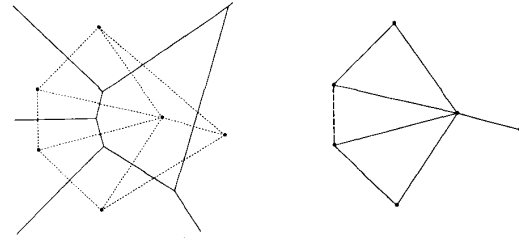


Fig. 3. $DT(V)$, the Voronoi diagram of V , and $GG(V)$.

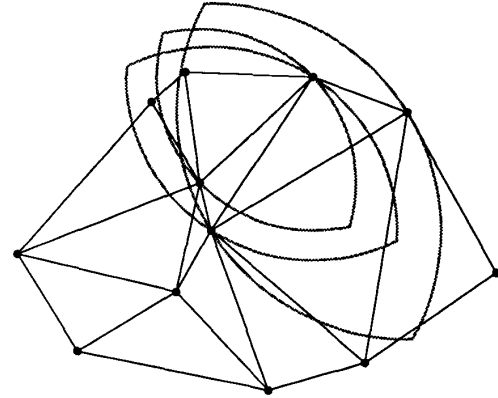


Fig. 4. $DT(V)$ and lunes showing an elimination path.

is eliminated by v ; the obtained sequence of edges will be called an elimination path of v ; see Fig. 4. This process does not necessarily detect all edges eliminated by v . Yet one can prove that each edge not in the *RNG* belongs to the elimination path for some point in V . If one continues constructing elimination paths for other points it will be noticed that paths that happen to coincide will never split apart. They grow into *elimination trees* (this term is slightly misleading since some elimination paths can have a cycle). This follows from an observation that if a point eliminates an edge of a triangle, then it may eliminate only the strictly longer of the two remaining edges of this triangle. This special structure allows an efficient elimination of edges not in the *RNG*. The elimination paths and elimination trees are built to form a so-called elimination forest, which carries all the information about the *RNG*. A *union-find* structure is helpful [46].

We can think about this construction as a sweeping guided by the shape of the $DT(V)$ rather than by the coordinate system. This sweeping can be accomplished in time $O(n)$ using the Gabow and Tarjan [46] implementation for a special case of *union-find* structure. If we are willing to sacrifice the linear time of this phase to gain simplicity, we can use another efficient and simultaneously very simple, almost linear, implementation for *union-find* such as a height ranking with path compression (see [47]).

It is interesting to note that the order of constructing elimination paths and composing them into an elimination forest is irrelevant. In general, different orders will lead

to different elimination forests; nevertheless all edges not in the *RNG* will always be eliminated. An elimination forest is an important structure and can be useful in other algorithms.

Recently, Hwang [48] used the idea of elimination paths to construct the *RNG* in a divide and conquer fashion. Elimination forests are carried over through the levels of recursion.

In view of linear expected time algorithms to construct the $DT(V)$ for some distributions of points (see [32], [49], [40], and [50]), the idea of elimination forests and the results of [15] give a simple $O(n)$ expected time algorithm for β -graphs.

We will finish this section with a simple algorithm to illustrate the use of the point location method in the elimination phase. This algorithm finds, for a planar point set, all edges in the *RNG* that are adjacent to a given vertex v . As we will see in the next section, the algorithm has a natural generalization to three-dimensional space.

We start with the set of all edges adjacent to the vertex v . A currently shortest edge e is found and all edges of the length equal to e are stored. Next all edges longer than e that form with e an angle less than $\frac{\pi}{3}$ are discarded; clearly they do not belong to the *RNG*. For the remaining (longer than e) edges the same process is repeated until no edges are left. At this moment edges of the same length are grouped into *orbits*. Clearly, by the construction, the number of orbits associated with v is not larger than $(2\pi)/(\frac{\pi}{3}) = 6$.

Not all edges in the orbits have to be in the *RNG* and the next phase will discard them. A helpful geometric fact is that if the angle between two adjacent edges e and w (w a shorter edge) is smaller than $\alpha_w = \arccos(\frac{|w|}{2|e|})$ then e does not belong to the *RNG*; $|e|$ is fixed for a given orbit. In other words, if for e there is a shorter edge w such that e is contained in the angular section (cone) with the axis w and the angle equal to $2\alpha_w$, then e is not in the *RNG*. In this case e is the longest edge in the triangle spanned by e and w . Fig. 5 illustrates this elimination. The edges on the orbit which intersect the perpendicular bisector of e are not in the *RNG*; each of them forms with w an angle less than α_w .

Now the problem reduces to the location of e in the union of angular sectors determined by all edges that are adjacent and shorter than e . After sorting the edges with respect to angles, such a union can be found in a linear time. In fact, edges in all orbits can be sorted in an $O(n^2)$ rather than $O(n^2 \log n)$ time. To this end observe that the order of the slopes of lines determined by n points can be found in $O(n^2)$ time (see [51]). A simple binary search determines in $O(\log n)$ steps whether e belongs to this union. Since there are at most six orbits for a given point and at most n edges, the total time to identify all edges in the *RNG* that are adjacent to a given point v is $O(n \log n)$. Clearly, the same result can be obtained using the Delaunay diagram. We will see in further sections that the above ideas can be extended to the three-dimensional Euclidean case.

In R^2 the edge location in the union of angular sectors, or equivalently a point location in the union of arcs,

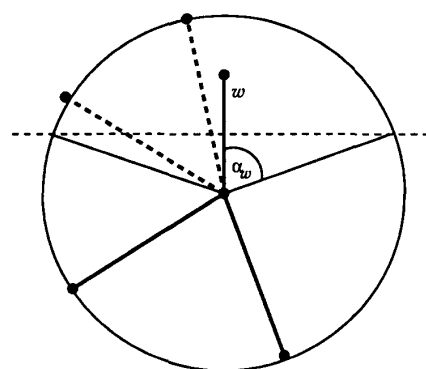


Fig 5. An orbit and its edges eliminated by w .

can be accomplished faster, in time proportional to n (as suggested by one of the referees.) Observe that each angle of elimination $2\alpha_w > \frac{2\pi}{3}$ and the corresponding arc occupy at least one third of a circle. Let a, b, c be equidistant points on the circle. We can group the arcs with respect to which of a, b, c they contain (ties can be resolved arbitrarily). The union of arcs in each group can be determined by finding the extremal arc endpoints in time proportional to the number of arcs. Now each point can be located in the union of the arcs in a constant time. Consequently, all edges in the *RNG* which are adjacent to a given point can be found in an $O(n)$ time.

For completeness, observe that this result yields immediately an $O(n^2)$ algorithm for the *RNG* in a Euclidean real plane. It is also possible to show that the total number of edges in all the orbits is subquadratic.

We will finish this section by mentioning two special cases.

Supowit [44] considered the *RNG* problem for convex polygons and he provided an optimal $O(n)$ time algorithm. Note that the same result can be established by combining a linear time construction of Delaunay triangulation for points forming a convex hull presented by Aggarwal *et al.* [52] with a linear time algorithm for constructing the *RNG* from the DT [15].

A linear time algorithm for computing the *RNG* of unimodal polygons, not necessarily convex, was designed by Olariu [53].

B. Three-Dimensional Space

A straightforward approach gives an $O(n^3)$ time algorithm for the *RNG* in three dimensions (see [1]).

In a manner similar to the two-dimensional case this time complexity can be reduced by constructing a "small" supergraph of the *RNG*. The first such construction was given by Supowit [44] for Euclidean spaces. The space around each point of $v \in V$ is partitioned into a number of narrow regions. Recall that a region is narrow if for $p, s \in V$ that belong to the region of v we have $\delta(p, s) < \max\{\delta(v, p), \delta(v, s)\}$. The number of regions is independent of n . For points in general position, based on such a

partition, it is easy to construct a linear size supergraph of the *RNG* in $O(n^2)$ time. Note that it can be also done in $O(n^2)$ time using the technique of Yao [34]. Specifically, the only candidate for the *RNG* in a given region is the edge joining v with the point in this region closest to v (the geographic neighbor). Because of the assumption about general position, in each region there is only a bounded (by a constant) number of such points. Note that the original assumption in [44] is that there are no isosceles triangles formed by the points in a given set. The algorithm works also if the number of equal length edges adjacent to each vertex is bounded by a common constant; in particular it holds for points in general position. A straightforward elimination of edges from this graph yields the *RNG* in total time $O(n^2)$. By virtue of the results obtained by Yao [34] and by Gabow, Bentley, and Tarjan [33], Supowit's algorithm extends to L_p metrics.

Elaborating on the above ideas and using balancing, Smith [26] showed how the time complexity can be reduced to $O(n^{23/12} \log n)$; general position is assumed.

Using another approach, Jaromczyk and Kowaluk [25] demonstrated a simple $O(n^2)$ algorithm for points in general position in L_p (the original assumption was a lack of isosceles triangles). For each point p , from the set of edges adjacent to p we remove iteratively the currently shortest edge pq and eliminate all edges pr such that $\delta(p, r) > \delta(q, r)$. Each such pr is the longest edge in the triangle Δpqr and therefore is not in the *RNG*. The *RNG* is the collection of edges which survive such a test, i.e., are never eliminated in this process. When this process terminates the number of edges adjacent to p is bounded by a constant, provided that there are no two equal distances between points of V (or their number is bounded by a constant). This follows from the fact that neither of $pw, pq \in \text{RNG}(V)$ is the longest edge in the triangle Δpwq and the angle opposite to the longest edge is greater than some constant $c(d, L_p) \times n$. Hence the overall cost of the algorithm is $O(c(d, L_p)n^2)$.

A similar method can be used for constructing the Gabriel graph of V in Euclidean spaces. A useful observation here is that $pr \notin GG(V)$ if there is q such that $\angle pqr > \pi/2$. Starting with the shortest edge pq adjacent to p , we eliminate all edges pr with $\angle pqr > \pi/2$. The process is iterated for all currently shortest noneliminated edges until no more edges adjacent to p are eliminated. The edges, E , which remain after such elimination are in the *GG*; the cost associated with each edge in E is proportional to n . It gives an $O(|E|n)$ algorithm for the *GG* in the Euclidean R^d space. The cost of the algorithm is $\Omega(n^2)$ since the *GG* is connected and has at least $n - 1$ edges. The algorithm does not depend on general position of input points. A variant of this algorithm is presented in [26].

A subquadratic algorithm for the *RNG* of points in d -dimensional spaces has been offered by Sun and Chang [54]. The algorithm constructs the *RNG*(V) of an n point set in a Euclidean d -dimensional space (d being fixed) in time $O(n^{2-a(d)}(\log n)^{1-a(d)})$, where $a(d) = 2^{-(d+1)}$. The points are assumed to be in general position. The

geographical nearest neighbor graph is used as a departure point. Recall that the *GNG* is a linear size supergraph of the *RNG* (see [44]), and it can be constructed in $O(n^{2-a(d)}(\log n)^{1-a(d)})$ (see [34]). The next stage, an elimination, uses an arrangement of $(d + 1)$ -dimensional hyperplanes. The hyperplanes are the images, in the inverse transformation, of those spheres which bound lunes generated by the edges of the *GNG*. Next the algorithm identifies lunes which contain a point from V . By well-known properties of the inverse transformation, the point location in an arrangement of spheres (lunes) in R^d is translated into a location problem in the corresponding arrangement of hyperplanes in R^{d+1} . Su and Chang used an algorithm from Dodge [55] and the balancing technique to preprocess hyperplanes for supporting point location queries.

Using faster existing algorithms for the nearest neighbor search in three-dimensional spaces they obtained an $O(n^{29/15} \log n)$ time algorithm to construct the *RNG* of points in R^3 (recall that general position is assumed).

Extending the idea of elimination based on angles between edges, Jaromczyk and Kowaluk [19] demonstrated an $O(n^2 \log n)$ time algorithm for arbitrary point sets in three-dimensional Euclidean space. The algorithm finds for each vertex v all edges of the *RNG* adjacent to v . As in the two-dimensional case, *orbits* of edges of the same length are formed (see the previous subsection). The number of orbits for each vertex is bounded by a constant independent of n . This constant is equal to the maximum number of segments with a common endpoint which form angles not less than $\frac{\pi}{3}$.

The orbits associated with v contain all edges in the *RNG* adjacent to v . However, they can also include extra edges. The size of the graph determined by the union of the orbits for all the vertices is $O(n^{3/2+\epsilon})$; see [19] or Section IV. A straightforward elimination leads to an $O(n^{5/2+\epsilon})$ algorithm. We can capitalize, however, on a special form of orbits and perform this elimination faster.

Recall that if the angle between two adjacent edges e and w (w a shorter edge) is smaller than $\arccos(\frac{|w|}{2|e|})$, then e does not belong to the *RNG*. This fact, which we have used to derive a two-dimensional algorithm, can be utilized in this situation as follows. Consider an orbit of v and let e belong to this orbit. For each edge w with an endpoint v which is shorter than e , take an open cone of revolution with its axis containing w and the angle at its apex equal to $2 \times \arccos(\frac{|w|}{2|e|})$. Clearly, e is not in the *RNG* if e is in the union of such cones (with v added).

Alternatively, we can use spherical cups obtained by intersecting the cones with the boundary of the sphere $B(v, |e|)$ and locate the endpoint of e in the union of these cups. A suitable stereographic projection transforms the problem to a point location in the union of (open) circles. To solve this problem, we can use Voronoi diagrams in Laguerre geometry [56] or power diagrams [57]. After $O(n \log n)$ preprocessing such a location can be done in $O(\log n)$ time per point. Hence for each vertex v the cost of eliminating extra edges from orbits is $O(n \log n)$, which gives in total $O(n^2 \log n)$ time.

Table 1 The Relative Neighborhood Graphs—Summary of Results

Dimension	Metric	Complexity	Reference	Remarks
R^2	L_p	n^2	[1], [41], [18]	[41] in L_2
		$n \log n$	[44]	in L_2
		n	[15]	when DT is given
	L_∞, L_1	$n^2 \log n$	[13]	
		$n \log n$	[45]	general position
		$n \log n + m$	[18]	m output size
R^3	L_2	$n^{5/2+\epsilon}$	[58]	
		$n^2 \log n$	[19]	
		$n^{29/15} \log n$	[54]	general position
		$n^{7/4+\epsilon}$	[22]	
		$n^{3/2+\epsilon}$	[22]	general position
	L_p	n^2	[44], [25], [58]	general position
	L_∞	$n^2 \log n$	[13]	general position
		$n(\log n)^2$	[26]	$n(\log n)^{d-1}$ in R^d

Katajainen and Nevalainen [58] designed a simple algorithm based on Urquhart [59]. The algorithm works in d -dimensional spaces and the metrics L_1, L_p, L_∞ . Interestingly, its time complexity analysis uses the region approach. In particular, they showed that in R^2 the running time is bounded by $O(n^{5/2})$. Using the methods of [19], we can in fact prove that this algorithm has $O(n^{5/2+\epsilon})$ time complexity for arbitrary point sets in three-dimensional Euclidean space.

Noticeable progress in three-dimensional and higher dimensional spaces has been recently made by Agarwal and Matoušek [22]. In the first step a family of well-separated pairs of subsets of V is constructed (see Section IV for the definitions of the well-separated pairs and bichromatic neighbors). This family has a property that each RNG edge connects some bichromatic closest neighbors in a separated pair of the family. Therefore, computation of all bichromatic closest neighbors for all the points in all the separated pairs gives a supergraph of the RNG . To eliminate from this supergraph edges which are not in the $RNG(V)$, an algorithm for a fast point location in an arrangement of lunes is designed. It uses a data structure based on the partitioning scheme of Chazelle *et al.* [60] and an arrangement of spheres based on the work of Clarkson *et al.* [61]. If points are in general position then: for a set V of n points in R^3 the $RNG(V)$ can be computed in $O(n^{3/2+\epsilon})$, for every $\epsilon > 0$. In R^d , for points in general position the $RNG(V)$ can be constructed in time $O(n^{2(1-\frac{1}{d+1}+\epsilon)})$.

It is also possible (see [22]) to construct the $RNG(V)$ in R^3 in time $O(n^{7/4+\epsilon})$, $\epsilon > 0$, for arbitrary point sets.

In the L_∞ metric the elimination phase can be based on fast point queries in d – rectangles (intersections of spheres in L_∞). In particular, in R^d the RNG can be constructed in $O(n(\log n)^{d-1})$ time; a general position of points is assumed (see [61]).

The results discussed in this section for the RNG are summarized in Table 1, where “big- O ” notion is omitted.

VI. VARIANTS AND SPECIAL CASES

The concept of relative neighborhood as a method to define graphs can be extended and modified in numerous directions. Usually modifications regard either elimination rules or the shape of a neighborhood. This section will discuss some of the interesting variants. The first group of modifications pertains to the relaxation of elimination rules.

A. k -Relative Neighborhood Graphs

Let $\Lambda_{p,q} = B(p, \delta(p, q)) \cap B(q, \delta(p, q))$; i.e., $\Lambda_{p,q}$ is a lune. The edges of the k -relative neighborhood graph of V , k – $RNG(V)$, are defined as follows:

$$(p, q) \in E \text{ if and only if the cardinality of } \Lambda_{p,q} \cap V \text{ is less than } k. \quad (6)$$

Clearly, the 1 – $RNG(V)$ is the $RNG(V)$; see Fig. 6. Properties and applications of k – $RNG(V)$ in Euclidean two-dimensional spaces are discussed in [62]. In general, k – RNG graphs in R^2 can contain intersecting edges yet their size is linear with respect to kn .

A simple probabilistic argument showing that the size of the k – RNG in R^3 is $O(k^{2/3}n^{4/3})$ was given in [22].

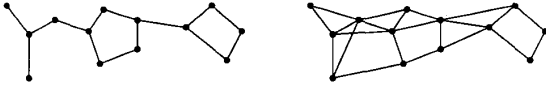


Fig. 6. RNG and $2-RNG$.

Consider a subset $R \subset V$ of $r = \lfloor \frac{n}{k} \rfloor$ points; each subset chosen with equal probability. A pair of points p, q which form an edge in the $k-RNG(V)$ are also an edge in the $RNG(R)$ if $p, q \in R$ and $\Delta_{p,q} \cup V$ is empty. The probability of such an event is at least

$$\frac{\binom{n-k-2}{r-2}}{\binom{n}{r}} = \Omega(1/k^2).$$

Therefore, the expected size of the $RNG(R)$ is bounded by $\Omega(M/k^2)$, where M is the size of the $k-RNG(V)$. However, in R^3 the size of the RNG of an $r = \lfloor n/k \rfloor$ element set is $O(r^{4/3})$ (see Section IV). Hence $M = O(k^{2/3}n^{4/3})$.

Clearly, a similar argument can be used for other k -neighborhood graphs. For example a bound of $O(kn)$ can be established for the $k-RNG$ in R^2 . Let us note that the same bound was also proved in [62] based on the size of the k -geographic neighbors graph.

An $O(n^{5/3} \log n)$ algorithm to construct the $k-RNG$, for a fixed k , has been presented in [62]. The algorithm has a traditional organization and works in two stages. At first the $k-GNG(V)$, the k -geographical neighborhood graph of V , is constructed. The $k-GNG$ is a generalization of the geographical neighborhood graph and is defined as follows. Let W be a narrow region. A point $p \in V \cap W$ is a k th nearest geographical neighbor of v in W if there are exactly $k-1$ points of V in W which are closer to v than p . The $k-GNG$ is obtained by connecting points with their first through k th nearest geographical neighbors. The k -geographical neighborhood graph is a supergraph of the $k-RNG$ and it can have at most $18kn$ edges (see [62]). The construction of the $k-GNG$ is modeled on an algorithm for the geographical neighborhood graph presented in [34].

In the second stage the lunes determined by edges in the $k-GNG$ are preprocessed for supporting fast point location queries. Using the balancing technique it is possible to preprocess lunes and perform all n point locations in the lunes, in order to eliminate edges which have at least k neighbors, in time not exceeding the cost spent on constructing $k-GNG$.

Recently, Agarwal and Matoušek [22] showed that the $k-RNG$ of a set of n points in R^d in general position, where k is a fixed constant, can be computed in time $O(n^{2(1-\frac{1}{d+1}+\epsilon)})$, $\epsilon > 0$. If points are in arbitrary position, then the $k-RNG$ can be computed in time $O(n^{4/3+\epsilon})$ for points in R^d , and in time $O(n^{7/4+\epsilon})$ for points in R^3 .

In Chang *et al.* [63] the $k-RNG$ is studied in an interesting context of bottleneck Hamiltonian cycles. For an application of the $k-RNG$ to Euclidean bottleneck matching and biconnected edge subgraphs, see [64] and [113].

A *minmax (bottleneck) Hamiltonian cycle* of a graph is a Hamiltonian cycle which minimizes the length of its longest edge over all Hamiltonian cycles of this graph. Consider a complete graph induced by a set V of points in a Euclidean plane. Such a graph has a minmax Hamiltonian cycle which is contained in the $20-RNG(V)$ (see [63]). In fact, there is a transformation which after a finite number of steps produces, from any given minmax Hamiltonian cycle, a minmax Hamiltonian cycle which is a subgraph of the $20-RNG$. Correctness of the transformation is based on an interesting observation related to narrow regions. Consider a set S of points contained in $\text{cone}(\vec{v}, \vec{u})$, $\angle(\vec{v}, \vec{u}) < \alpha$ and t the apex of the cone. Assume that for all points $p, q \in S$ the following properties hold, $r > 0$: $\delta(t, p) \geq r$, $\delta(p, q) \geq r$, $\delta(p, q) \geq \delta(t, p) - r$, and $\delta(p, q) \geq \delta(t, q) - r$. Then there exists a constant c depending only on α such that $|S| < c$ (see [63]).

Now consider a minmax Hamiltonian cycle H and its longest edge pq not in the $k-RNG$. (The value of k can be specified; $k = 20$ in [63].) Since pq is not an edge in the $k-RNG(V)$, the lune $\Delta_{p,q}$ contains at least k points, $t_1, \dots, t_m$, of V . If any pair t_i, t_j is connected by an edge in H , then we can replace pq by a sequence $pt_i, t_i t_j, t_j q$; the new cycle is again a minmax Hamiltonian cycle. Otherwise, let w_i be a direct predecessor of t_i in H ; assume that H has a counterclockwise orientation. If k is large enough, then by virtue of the above observation on the narrow regions there are two points w_i, w_j that satisfy at least one of the following conditions:

- a) $\delta(q, w_i) < \delta(p, q)$, or
- b) $\delta(w_i, w_j) < \delta(p, q)$, or
- c) $\delta(w_i, w_j) < \delta(w_i, t_i)$ or $\delta(w_i, w_j) < \delta(w_j, t_j)$.

In each case the edge pq can be replaced by other edges without increasing the length of the longest edge. For example, in case (b) the edges $pq, w_i t_i, w_j t_j$ can be substituted with $pt_i, t_i t_j, w_j q$. Moreover, each application of this transformation replaces a longest edge not in the $k-RNG$ by some shorter edges. Hence, after a finite number of steps we obtain a minmax Hamiltonian cycle which is a subgraph of the $k-RNG$.

A related and important problem of constructing a *minmax length triangulation*, i.e., a triangulation which minimizes the longest edge, has been studied by Edelsbrunner and Tan [65]. They have proved that every finite point set V in R^2 has a minmax length triangulation which is a supergraph of the $RNG(V)$. The lemma suggests that construction of a minmax length triangulation can start from the $RNG(V)$. It leads immediately to a cubic time construction when existing dynamic programming algorithms for triangulations are utilized (see [66]). It also provides the first polynomial time algorithm for this problem. Furthermore, as demonstrated in [65], an even faster quadratic time algorithm can be developed. The algorithm works for a general class of metrics.

B. k -Gabriel Graphs

In a similar fashion k -Gabriel graphs can be introduced

(see [67]). Let $\Gamma_{p,q} = B(\frac{p+q}{2}, \frac{\delta(p,q)}{2})$; i.e. $\Gamma_{p,q}$ is a diameter sphere. The edges of the k -Gabriel graph of V , the k -GG(V), are defined as follows:

$$(p, q) \in E \text{ if and only if the cardinality of } \Gamma_{p,q} \cap V \text{ is less than } k. \quad (8)$$

Properties of these graphs and applications to bottleneck biconnected graphs, and Euclidean bottleneck matchings are discussed in [67], where an $O(k^2 n \log n)$ time construction for the k -GG in R^2 is presented.

C. Constrained Relative Neighborhood Graphs and Gabriel Graphs

An interesting extension of relative neighborhood graphs has been investigated by Su and Chang [68]. Let V be a set of n points in a plane and T be a set of nonintersecting line segments with their endpoints in V . Clearly, the number of segments in T is of order $O(n)$. We say that two points in V are visible if their connecting line segment does not intersect any edge in T .

The *constrained relative neighborhood graph* of $V \cup T$, $CRNG(V \cup T)$, is defined as a graph with vertices V and the set of edges E such that $(p, q) \in E$ if and only if:

- 1) $(p, q) \in T$, or
- 2) p, q are visible and $\Lambda_{p,q}$ does not contain points in V which are visible from both p and q .

Figure 7 illustrates a polygon T , the $RNG(V)$ of its vertices, and its $CRNG$. Interestingly, a special case of the $CRNG$, where T forms a simple polygon, was introduced much earlier in [69]. Because of its applications to pattern recognition we will discuss this special case separately.

It appears that $CRNG(V \cup T)$ is a subgraph of the *constrained Delaunay triangulation* of V and T (see [68]). The constrained Delaunay triangulation of V and T , $CDT(V, T)$, contains all segments in T . In addition, it includes all edges (p, q) such that p, q are visible and there is a sphere with its boundary containing p, q which does not contain points of V visible from both p and q . The $CDT(V, T)$ can be constructed in $O(n \log n)$ time (see [12]). Using a process similar to Supowit's sweep, it is possible to eliminate edges from $CDT(V, T) \setminus CRNG(V \cup T)$ in $O(n \log n)$ time.

Constrained Gabriel graphs, CGG , are defined in an analogous fashion. They can be obtained from the CDT in $O(n)$ time. To this end it is sufficient to consider obtuse and right angle triangles in the CDT . After eliminating from these triangles the longest edges which are not in T we obtain the desired CGG [68].

Clearly, the whole concept can be extended to β graphs. Furthermore, using the elimination forest structure of [15], the constrained β graph ($1 \leq \beta \leq 2$) can be constructed in an $O(n)$ time.

D. Relative Neighborhood Decomposition of a Simple Polygon

The relative neighborhood decomposition of a simple polygon is connected to the constrained relative neighbor-

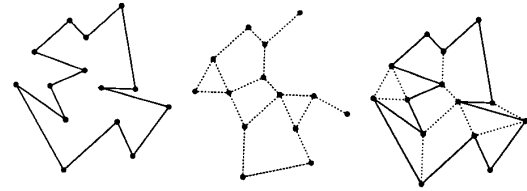


Fig. 7. A polygon, the RNG of its vertices, and its $CRNG$.

hood graphs discussed earlier in this section. Let P be a simple polygon and V be its set of vertices. We set the neighborhood $U_{p,q}$ to be the lune $\Lambda_{p,q}$.

The relative neighborhood decomposition of P , $RND(P)$, is defined as a set of these edges pq which are diagonals of P , and if $v \in \Lambda_{p,q}$ then at least one of pv, qv intersects the boundary of P .

The concept of the RND has been introduced by Toussaint [70]. Intuitively, we can think that an edge of P is an opaque barrier. A point can have an impact on the segment pq only if no barrier separates them.

The RND can be used to decompose simple polygons into perceptually meaningful components and is thus useful in pattern recognition.

ElGindy and Toussaint [69] discussed properties and several algorithms for the RND as well as the Gabriel decomposition, which is defined analogously. In particular, they show that the $RND(P)$ is a planar graph and that it can be constructed in time $O(n^2)$ (n being the number of edges in P).

The RND can be viewed as a special case of a constrained relative neighborhood graph. Based on algorithms for the $CRNG$, it can be constructed in time $O(n \log n)$.

E. The Rectangular Influence Graph

Ichino and Sklansky [71] defined the rectangular influence graph, $RIG(V)$, assuming as the neighborhood $U_{p,q}$ of points p, q the smallest coordinate oriented rectangle containing these points. They showed that the $RIG(V)$ is a supergraph of $GG(V)$ and therefore also a supergraph of the $RNG(V)$ in L_p . However, in general, the rectangular influence graph is not a subgraph of the DT .

In addition, variants such as an interclass RIG and mutual neighborhood graphs were introduced. In our terminology, the mutual neighborhood graph can be viewed as a graph defined over two point sets V_1 and V_2 . The mutual graph $MNG(V_1|V_2)$ has edges for these pairs of points in V_1 , which neighborhood does not contain points from V_2 .

Several applications of the RIG , the interclass RIG , and the MNG are presented in [71]. In particular, it is argued that these graphs are useful in the design of piecewise-linear classifiers and in clustering methods applicable to mixed feature variables.

In this context we also mention direct dominance pairs, which are applicable to the rectangle enclosure problem. Algorithms for reporting all direct dominance pairs are given in the work by Gutting, Nurmi, and Ottman [72].

F. Other Graphs

There is a large family of neighborhood graphs that we have not mentioned here.

This family includes such important and interesting graphs as the *sphere of influence graph*, introduced by Toussaint [73] and α graphs introduced by Edelsbrunner, Kirkpatrick, and Seidel [74]. Both graphs are useful in *dot pattern* and *shape analysis* [75]. They also have several interesting graph-theory properties [76], [73], [74]. We expand on these graphs in Section VII.

Several variants of β graphs are discussed in [9]. The ideas in [9] can be generalized and expanded on in various directions. Some discussion of possible generalizations, in particular with an eye toward applications, is given in [75].

A recent paper by Veltkamp [77] proposes a parametrized family of γ -neighborhood graphs. The γ graph can be reduced to the circle based β skeleton, to the *DT*, or to the convex hull for specific parameter values.

An interesting variant of a *digital geometry* version of the *RNG* applied to computer vision is given in the work by Toriwaki and Yokoi [78]. Planar neighborhood graphs without cycles are discussed in [114].

VII. APPLICATIONS

We have already touched briefly on some applications as different neighborhood graphs were introduced. In this section we consider applications in more detail. We will discuss graphs and structures which are related to the *RNG* as members of the same family belonging to computational morphology.

A. Morphology and Computer Vision

1) *Dot Patterns*: The central problem in computer vision begins with a gray-level intensity array of a visual pattern or scene and culminates with a description of the scene. This description is a marriage of the intensity array itself (the bottom-up or data-driven information) and the objective for which it is viewed (the top-down or conceptually-driven information). Considerable attention has been devoted to *low-level* vision, i.e., the aspect dealing with the analysis of the intensity array up to the level of figure-background separation. The main purpose of analyzing an intensity array at this level is to form a low-level description that is independent of any final conceptually-driven description. Marr [79] has called such a description a primal sketch of the intensity array. One class of patterns or scenes which had been studied extensively is the so-called dot pattern.

When dots in the plane have a finite diameter so that they are visible, and when they are fairly densely distributed in some region in the plane, then a human observer is quick to perceive the "shape" of such a set. These sets are usually referred to as dot patterns or dot figures. A polygonal description of the boundary of the shape is referred to as the shape hull of a dot pattern, where the vertices are given as the coordinates of the dot centers.

There are two versions of the shape hull problem. In one there are no "holes" in the dot pattern and the dot

pattern is "simply connected"; hence the shape hull is a simple polygon. On the other hand, in the more difficult problem both "holes" and "disconnected" components may exist. To add to this difficulty, in some instances illusory contours are perceived between "disconnected" components as illustrated by Kennedy and Ware [80]. For more details and early approaches see Toussaint [81] and Medek [82].

In addition to describing the shape or structure of a set of points by its shape-hull or external shape we may also use the *skeleton* or *internal shape*. An early step in this direction was taken by Zahn [83] with the minimal spanning tree. More recent approaches have used the *RNG* [71].

Most of the early approaches suffer from various deficiencies such as computational inefficiency or dependence on too many parameters which must be fine-tuned in order to obtain satisfactory performance for the task at hand.

A very elegant definition for the external shape of a set of points was put forward by Edelsbrunner, Kirkpatrick, and Seidel [74]. They have proposed a natural generalization of convex hulls that they call α hulls. The α hull of a point set is based on the notion of generalized disks in the plane. For a real number α a generalized disk of radius $\frac{1}{\alpha}$ is a (standard) disk of radius $\frac{1}{\alpha}$ if $\alpha > 0$. If $\alpha < 0$ it is the complement of a disk of radius $\frac{1}{\alpha}$. It is half-plane for $\alpha = 0$.

The α hull of a point set S is defined to be the intersection of all closed generalized disks of radius $\frac{1}{\alpha}$ that contain all the points of S . The convex hull of S is exactly the 0 hull. The family of α hulls includes a spectrum of enclosing regions of S .

A combinatorial variant of the α hull defined in [74] is called the α shape. It can be viewed as the boundary of the α hull with curved edges replaced by straight edges. Unlike the family of α hulls, the family of distinct α shapes has only a finite number of members. They provide a spectrum of progressively more detailed descriptions of the external shape of a given point set; they can be calculated in $O(n \log n)$ time for n point sets. For more details see [74] and [11].

Note that the idea of α hulls is closely related to the notions of *opening* and *closing* sets, found in mathematical morphology (see [84]–[86]). The two-dimensional α shapes are connected to the dot patterns (see [87] and [88]) and to the circle diagrams utilized in cluster analysis (see [89]). An extension of α shapes to R^3 and an $O(n^2)$ algorithm that constructs for n points a representation of the α shapes for all values of α have been presented by Edelsbrunner and Mücke [90]. The shape hulls are also used in muscle fiber analysis [115]. A digital α -hull introduced by Saito *et al.* [116] was used for shape feature analysis.

A new methodology for describing the *internal shape* of point sets was outlined by Kirkpatrick and Radke [9], where the notion of β graphs was introduced.

Another simple and elegant method for extracting the shape of a dot pattern is due to Rosenberg and Langridge [91]. The algorithm is free of parameter tuning. It is limited, however, to single objects; its computational complexity is $O(n^4)$.

Toussaint [73] proposed a graph that appears to cap-

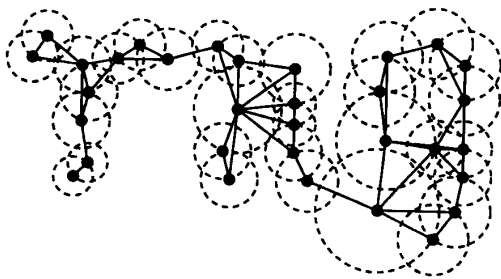


Fig. 8. The sphere-of-influence graph.

ture the essence of the *primal sketch* for dot patterns of arbitrary complexity. It seems that the graph, which is called the sphere-of-influence graph, suffers from none of the drawbacks of the previous methods. It delivers either the “internal” structure in the form of a “skeleton” or the “external” structure in the form of a “shape-hull” as a function of what the data look like (see Fig. 8). Secondly, it can be applied to a scene of disconnected objects and it works without tuning of parameters. In addition, the graph affords a graph-theory explanation of certain visual illusions, such as the Mueller-Lyer illusion.

Let $V = \{p_1, \dots, p_n\}$ be a finite set of points in the plane. For each point $p \in V$, let r_p be the closest distance to any other point in the set, and let C_p be the circle of radius r_p centered at p . The sphere of influence graph is a graph on V with an edge between points p, q if and only if the circles C_p, C_q intersect in at least two places. The sphere of influence graph has at most cn edges, where c is a constant (see [76]). This constant is not greater than $17\frac{1}{2}$; for a discussion of this issue see [76] and [92], as well as [93] and [94]. It implies, as was observed by ElGindy, that the algorithm of Bentley and Ottman [95] for reporting intersections can be used to find the sphere of influence graph in $O(n \log n)$ time. This is optimal in the decision tree model (see [76]).

A general family of graphs related to the sphere-of-influence graph, although based on a slightly different definition and with a different purpose, has been studied in [92] and [96].

Guibas, Pach, and Sharir [97] have recently generalized the sphere-of-influence graph to *kth sphere of influence*. Given a set V of n points in R^d , the *kth sphere of influence* of a point $x \in V$ is the smallest closed ball centered at x and containing more than k points of V (including x). The case $k = 1$ gives the standard sphere of influence. The *kth sphere-of-influence graph* $G_k(V)$ of V is a graph whose vertices are the points of V , and two points are connected by an edge if their *kth* spheres of influence intersect. They show, extending results for the first sphere-of-influence graph from [76] and [92], that there is a constant $c_d > 0$ depending only on d such that the number of edges in $G_k(v)$ is at most $c_d kn$. They also give an algorithm that computes the *kth* sphere-of-influence graph in time $O(n^{2 - \frac{1}{1 + \lceil \frac{d+2}{2} \rceil} + \epsilon} + kn \log^2 n)$ for any $\epsilon > 0$.

Even for $k = 1$ the sphere-of-influence graph need be neither connected nor planar. In [98] trees that are sphere-of-influence graphs are characterized. The geometric notion of the sphere of influence has also been modified to a graph theory sense by Harary *et al.* [99].

2) *Texture Discrimination*: There are many computer vision problems where the patterns are neither dot patterns nor line patterns but rather *textures*. This is particularly so in satellite image analyses of the earth.

Toriwaki and Yokoi [78] have applied the *RNG* and the *GG* to the problem of discriminating between different textures.

3) *Monotonic Search Networks*: A graph embedded in the plane is called monotone if for every pair of vertices p, q there is a path $p = v_i, v_2, \dots, v_{k-1}, v_k = q$ such that $\delta(v_i, q) > \delta(v_{i+1}, q)$ for $i = 1, \dots, k-1$.

A useful property of monotone graphs is that the distance can be used to guide efficient traversal of the graph. Dearholt *et al.* [100] have used a monotone graph called the monotonic search network as the underlying structure for an associative data base for computer vision. Kurup [101] showed that the Delaunay triangulation is a monotone graph which in general is not true for the *GG*, the *RNG*, and the minimum spanning tree. He observed, however, that the *RNG* can be extended to a monotone graph, which is minimal in some sense, and consequently it can be used as a base for monotonic search networks [102], which find application in the design of data bases for computer vision.

B. Geographic Analysis

Underlying the study of geographic analysis is *spatial analysis* [29] the field of study which examines the spatial structure and association of phenomena. This is a large and well-established area of geography. The *RNG* and other proximity graphs may revolutionize the manner in which the spatial analysts do their work. For a survey of applications of neighborhood graphs to spatial analysis the reader is referred to the work by Radke [75].

C. Pattern Classification

In the *nonparametric classification problem*, a set of n feature vectors is available. It is taken from a collected data set: $\{X, \Theta\} = \{(X_1, \Theta_1), \dots, (X_n, \Theta_n)\}$, where X_i and Θ_i denote the feature vector of the *i*th object and the class label of this object, respectively. One of several powerful classification techniques is the so-called nearest-neighborhood rule (NN rule) [103], [104]. Let Y be a new object (vector feature) to be classified. The NN rule classified Y as belonging to class Θ_k , where $X_k \in \{X_1, \dots, X_n\}$ is the feature vector closest to Y .

In the past some practitioners avoided using the NN rule because early algorithms were inefficient and had to store all data $\{X, \Theta\}$. Both of these problems have been eradicated with techniques from computational geometry. Various methods exist for finding nearest neighbors including an optimal $O(n \log n)$ algorithm [105]. For other approaches see also [106] and [107]. Furthermore, not all

the "training" data $\{X, \Theta\}$ are required to be stored. Methods have been developed [35] to edit redundant members of $\{X, \Theta\}$ in order to obtain a relatively small subset of $\{X, \Theta\}$ that implements exactly the same decision rule as using all of $\{X, \Theta\}$. Such methods depend heavily on the use of Voronoi diagrams and proximity graphs such as the *RNG* and the Gabriel graph [35].

Let us finish with a few further applications of the *RNG*. Lee [108] applied the *RNG* to computing shortest rectilinear paths; it is an element of a design problem in *VLSI*. In Lefkovich [109] the *RNG* are used in cluster analysis in ecology. Finally, in [110] and [111] the *RNG* found an application in comparing dissimilarity matrices.

VIII. CONCLUSION

We have reviewed in this paper results and algorithms for neighborhood graphs. Although we have focused on the relative neighborhood graphs the other members of this rich family of graphs have been discussed as well.

Many interesting questions remain open. Among them there is a problem of tight bounds for the number of edges of the *RNG* in R^3 . The best established upper bound is superlinear. On the other hand only a trivial linear lower bound is known. In this context it would be interesting to develop optimal or output sensitive algorithms for the three-dimensional *RNG* as well as the *GG*. Also, three-dimensional β skeletons deserve more research. Another area which needs further study is the question of recognition of proximity graphs, i.e., given a class of proximity graphs and a graph G , determine if G belongs to this class. The known results concern the Delaunay triangulation [112] and f -factors of point sets in the plane [92].

In view of the widespread applications of neighborhood graphs to computational morphology, geographical analysis, and pattern analysis, the design of robust algorithms and implementations for the neighborhood graphs is a challenging problem.

ACKNOWLEDGMENT

The authors would like to thank the anonymous referees, whose valuable remarks were essential in improving the presentation of our paper. The first author would like also to thank K. Senn for her professional help in locating many of the references.

REFERENCES

- [1] G. T. Toussaint, "The relative neighborhood graph of a finite planar set," *Pattern Recognition*, vol. 12, pp. 261–268, 1980.
- [2] G. T. Toussaint, "Some unsolved problems on proximity graphs," in *Proceedings of the First Workshop on Proximity Graphs* (New Mexico State University, Las Cruces), 1991.
- [3] D. W. Dearholt and F. Harary, Eds., *Proceedings of the First Workshop on Proximity Graphs*. Computing Research Laboratory, New Mexico State University, 1991. Memoranda in Computer and Cognitive Science MCCC-91-224.
- [4] J. W. Boland, R. C. Brigham, and R. D. Dutton, "The difference between a neighborhood graph and a wheel," *Congressus Numerantium*, vol. 58, pp. 151–156, 1987.
- [5] W. Dunham, *Journey Through Genius: The Great Theorems of Mathematics*. New York: Wiley, 1990.
- [6] R. Lawlor, *Sacred Geometry: Philosophy and Practice*. London: Thames and Hudson, 1982.
- [7] P. M. Lankford, "Regionalization: Theory and alternative algorithms," *Geogr. Annals*, vol. 1, pp. 196–212, 1969.
- [8] K. R. Gabriel and R. R. Sokal, "A new statistical approach to geographic variation analysis," *Systematic Zoology*, vol. 18, pp. 259–278, 1969.
- [9] D. G. Kirkpatrick and J. D. Radke, "A framework for computational morphology," in *Computational Geometry*, G. Toussaint, Ed., North-Holland, 1985, pp. 217–248.
- [10] F. Preparata and M. I. Shamos, *Computational Geometry – Introduction*. New York: Springer-Verlag, 1986.
- [11] H. Edelsbrunner, *Algorithms in Combinatorial Geometry*. New York: Springer-Verlag, 1987.
- [12] F. Aurenhammer, "Voronoi diagrams — A survey of fundamental geometric data structure," *ACM Computing Surveys*, vol. 23, pp. 345–406, 1991.
- [13] J. O'Rourke, "Computing the relative neighborhood graph in the l_1 and l_∞ metrics," *Pattern Recognition*, pp. 45–55, 1982.
- [14] D. T. Lee, "Two dimensional Voronoi diagram in the l_1 -metric," *J. Ass. Comput. Mach.*, vol. 27, pp. 604–618, 1980.
- [15] J. W. Jaromczyk, M. Kowaluk, and F. Yao, "An optimal algorithm for constructing β -skeletons in l_p metric," *SIAM J. Comput.*, to be published.
- [16] D. W. Matula and R. R. Sokal, "Properties of Gabriel graphs relevant to geographical variation research and the clustering of points in the plane," *Geographical Analysis*, vol. 12, pp. 205–222, 1984.
- [17] R. B. Urquhart, "Some properties of the planar Euclidean relative neighborhood graph," *Pattern Recognition Lett.*, pp. 317–322, 1983.
- [18] J. Katajainen, "The region approach for computing relative neighborhood graphs in the l_p metric," *Computing*, vol. 40, pp. 147–161, 1988.
- [19] J. W. Jaromczyk and M. Kowaluk, "Constructing the relative neighborhood graph in 3-dimensional Euclidean space," *Discrete Applied Math.*, vol. 31, pp. 181–192, 1991.
- [20] K. L. Clarkson, H. Edelsbrunner, L. J. Guibas, M. Sharir, and E. Welzl, "Combinatorial complexity bounds for arrangements of curves and surfaces," *Discrete and Computational Geometry*, vol. 5, pp. 99–160, 1990.
- [21] F. R. K. Chung, "Sphere-and-point incidence relations in high dimensions with applications to unit distances and furthest-neighbor pairs," *Discrete and Computational Geometry*, vol. 4, pp. 183–190, 1989.
- [22] P. K. Agarwal and J. Matoušek, "Relative neighborhood graphs in three dimensions," in *Proc. 3rd Annual ACM-SIAM Symp. Discrete Algorithms*, 1992, pp. 58–67; also in *Computational Geometry: Theory and Applications*, to be published.
- [23] P. Agarwal, H. Edelsbrunner, O. Schwarzkopf, and E. Welzl, "Euclidean minimum spanning trees and bichromatic closest pairs" in *Proc. 6th ACM Symp. Computational Geometry*, 1990, pp. 203–210.
- [24] H. Edelsbrunner and M. Sharir, "A hyperplane incidence problem with applications to counting distances," in *Proc. Int. Symp. Algorithms, LNCS 450*, 1990, pp. 419–428.
- [25] J. W. Jaromczyk and M. Kowaluk, "A note on relative neighborhood graphs," in *Proc. 3rd Ann. Symp. Computational Geometry* (Waterloo, Canada), June 8–10, 1987, pp. 233–241.
- [26] W. D. Smith, "Studies in computational geometry motivated by mesh generation," Ph.D. thesis, Princeton University, 1989.
- [27] B. Chazelle et al., "Slimming down by adding: Selecting heavily covered points," in *Proc. 6th Ann. Symp. Computational Geometry*, 1990, pp. 116–127.
- [28] L. Devroye, "The expected size of some graphs in computational geometry," *Computers and Mathematics with Applications*, vol. 15, pp. 53–64, 1988.
- [29] A. Getis and B. N. Boots, *Models and Spatial Process*. Cambridge, U.K.: Cambridge University Press, 1978.
- [30] R. E. Miles, "On the homogenous planar poisson point process," *Mathematical Biosciences*, vol. 6, pp. 85–127, 1970.
- [31] B. M. Chazelle, "Filtering search: A new approach to query answering," *SIAM J. Comput.*, vol. 15, pp. 703–724, 1986.
- [32] J. Katajainen, "Bucketing and filtering in computational geometry," Ph.D. thesis, University of Turku, Finland, 1987. (Available as Report A46.)
- [33] H. N. Gabow, J. L. Bentley, and R. E. Tarjan, "Scaling and related techniques for geometry problems," in *Proc. 16th Ann. ACM Symp. Theory of Computing*, 1984, pp. 135–143.

- [34] A. C.-C. Yao, "On constructing minimum spanning trees in k -dimensional spaces and related problems," *SIAM J. Comput.*, vol. 11, pp. 721-736, 1982.
- [35] G. T. Toussaint, B. K. Bhattacharya, and R. S. Poulsen, "The application of Voronoi diagrams to non-parametric decision rules," in *Proc. Computer Science and Statistics: 16 Symp. Interface* (Atlanta), Mar. 14-16 1984.
- [36] D. E. Willard, "New data structures for orthogonal range queries," *SIAM J. Comput.*, vol. 14, pp. 232-253, 1985.
- [37] H. Edelsbrunner and M. H. Overmars, "Batched dynamic solutions to decomposable searching problems," *J. Algorithms*, vol. 6, pp. 515-542, 1985.
- [38] L. J. Guibas and J. Stolfi, "On computing all north-east nearest neighbors in the l_1 metric," *Inform. Process. Lett.*, vol. 17, pp. 219-223, 1983.
- [39] Y. C. Wee, "On the angle restricted nearest neighbor problem," Tech. Rep. 89-8, Dept. Comp. Sci., University at Albany, 1989.
- [40] J. L. Bentley, B. W. Weide, and A. C. Yao, "Optimal expected-time algorithms for closest point problems," *ACM Trans. Math. Software*, pp. 563-580, 1980.
- [41] J. Katajainen and O. Nevalainen, "Computing relative neighborhood graphs in the plane," *Pattern Recognition*, vol. 19, pp. 221-228, 1986.
- [42] J. Katajainen, O. Nevalainen, and J. Teuhola, "A linear expected-time algorithm for computing planar relative neighborhood graphs," *Inform. Process. Lett.*, vol. 25, pp. 77-86, 1987.
- [43] G. T. Toussaint and R. Menard, "Fast algorithms for computing the planar relative neighborhood graph," in *Proc. 5th Symp. Operations Research*, 1980, pp. 425-428.
- [44] K. J. Supowit, "The relative neighborhood graph, with an application to minimum spanning trees," *J. Assoc. Comput. Mach.*, vol. 30, pp. 428-448, 1983.
- [45] D. T. Lee, "Relative neighborhood graphs in the l_1 -metric," *Pattern Recognition*, vol. 18, pp. 327-332, 1985.
- [46] H. N. Gabow and R. E. Tarjan, "A linear-time algorithm for a special case of disjoint set union," *J. Computer and System Sciences*, vol. 30, pp. 209-221, 1985.
- [47] R. E. Tarjan, *Data Structures and Network Algorithms*. Philadelphia, PA: SIAM, 1983.
- [48] N.-F. Hwang, "Divide and conquer algorithm for RNG," *BIT*, vol. 30, pp. 196-206, 1990.
- [49] R. Dwyer, "Average-case analysis of algorithms for convex hulls and Voronoi diagrams," Ph.D. thesis, Carnegie-Mellon University, 1988.
- [50] A. Maus, "Delaunay triangulation and the convex hull of points in expected linear time," *BIT*, vol. 24, pp. 151-163, 1984.
- [51] M. H. Overmars and E. Welzl, "New methods for computing visibility graphs," in *Proc. Fourth Ann. Symp. Comp. Geometry*, 1988, pp. 164-171.
- [52] A. Aggarwal, L. J. Guibas, J. Saxe, and P. W. Shor, "A linear time algorithm for computing the Voronoi diagram of a convex polygon," in *Proc. 19th Ann. ACM Symp. Theory of Computing*, 1987, pp. 39-45.
- [53] S. Olariu, "A simple linear-time algorithm for computing the rng and mst of unimodal polygons," *Inform. Process. Lett.*, vol. 31, pp. 243-248, 1989.
- [54] T.-H. Su and R.-Ch. Chang, "On constructing relative neighborhood graphs in Euclidean k -dimensional spaces," *Computing*, vol. 46, pp. 121-130, 1991.
- [55] C. W. Dodge, *Euclidean Geometry and Transformations*. Reading, MA: Addison-Wesley, 1972.
- [56] H. Imai, M. Iri, and K. Murota, "Voronoi diagram in the Laguerre geometry and its applications," *SIAM J. Comput.*, vol. 14, pp. 93-105, 1985.
- [57] F. Aurenhammer, "Improved algorithms for discs and balls using power diagrams," *J. Algorithms*, vol. 9, pp. 151-161, 1988.
- [58] J. Katajainen and O. Nevalainen, "An almost naive algorithm for finding relative neighborhood graphs in l_p metrics," *Informatique Theoretique et Applications*, vol. 21, pp. 199-215, 1987.
- [59] R. B. Urquhart, "Algorithms for computation of relative neighborhood graph," *Electron. Lett.*, vol. 14, no. 14, pp. 556-557, 1980.
- [60] B. Chazelle, M. Sharir, and E. Welzl, "Quasi-optimal upper bounds for simplex range searching and new zone theorem," in *Proc. 6th ACM Symp. Computational Geometry*, 1990, pp. 23-33.
- [61] K. L. Clarkson, H. Edelsbrunner, L. J. Guibas, M. Sharir, and E. Welzl, "Combinatorial complexity bounds for arrangements of curves and surfaces," in *Proc. 29th Ann. Symp. Foundations of Computer Science*, 1989, pp. 568-579.
- [62] T.-H. Su and R.-Ch. Chang, "Computing the k -relative neighborhood graphs in Euclidean plane," *Pattern Recognition*, vol. 24, pp. 231-239, 1991.
- [63] M. S. Chang, C. Y. Tang, and R. C. T. Lee, "20-relative neighborhood graphs are Hamiltonian," in *Proc. Int. Symp. SIGAL '90*, Aug. 1990, pp. 53-65.
- [64] M. Chang, C. Tang, and R. Lee, "Solving the Euclidean bottleneck matching problem by k -relative neighborhood graph," *Algorithmica*, 1991.
- [65] H. Edelsbrunner and T. S. Tan, "A quadratic time algorithm for the minmax length triangulation," Tech. Rep., Dept. of Comp. Sci., University of Illinois at Urbana Champaign, Feb. 1991.
- [66] G. T. Klinscek, "Minimal triangulations of polygonal domains," *Annals Discrete Math.*, pp. 121-123, 1980.
- [67] T.-H. Su and R.-Ch. Chang, "The k -Gabriel graphs and their applications," in *Proc. Int. Symp., SIGAL '90*, Aug. 1990, pp. 66-75.
- [68] T.-H. Su and R.-Ch. Chang, "Computing the constrained relative neighborhood graphs and constrained Gabriel graphs in Euclidean plane," *Pattern Recognition*, vol. 24, pp. 221-230, 1991.
- [69] H. A. ElGindy and G. T. Toussaint, "Computing the relative neighborhood decomposition of a simple polygon," *Computational Morphology*, pp. 53-70, 1988.
- [70] G. T. Toussaint, "Decomposing a simple polygon with the relative neighborhood graph," in *Proc. Allerton Conf.*, Oct. 1980, pp. 20-28.
- [71] M. Ichino and J. Sklansky, "The relative neighborhood graph for mixed feature variables," *Pattern Recognition*, vol. 18, pp. 161-167, 1985.
- [72] R. H. Gutting, O. Nurmi, and T. Ottmann, "The direct dominance problem," in *Proc. First Ann. ACM Symp. Computational Geometry*, 1985, pp. 81-88.
- [73] G. T. Toussaint, "A graph-theoretical primal sketch," in *Computational Morphology*, G. T. Toussaint, Ed. North-Holland, 1988, pp. 229-260.
- [74] H. Edelsbrunner, D. Kirkpatrick, and R. Seidel, "On the shape of a set of points in the plane," *IEEE Trans. Inform. Theory*, vol. 29, pp. 551-559, 1983.
- [75] J. D. Radke, "On the shape of a set of points," in *Computational Morphology*, G. T. Toussaint, Ed. North-Holland, 1988, pp. 105-136.
- [76] D. Avis and J. Horton, "Remarks on the sphere of influence graph," in *Discrete Geometry and Convexity*, J. E. Goodman et al., Eds. New York Academy of Sciences, 1985, pp. 323-327.
- [77] R. C. Veltkamp, "The γ -neighborhood graph," Tech. Rep., Centre for Mathematics and Computer Science, July 1990.
- [78] J. I. Toriwaki and S. Yokoi, "Voronoi and related neighbors on digitized two dimensional space with application to texture analysis," in *Computational Morphology*, G. T. Toussaint, Ed. North-Holland, 1988, pp. 207-228.
- [79] D. Marr, "Early processing for of visual information," *Philosophical Trans. Royal Soc. London*, vol. 275, no. 945, pp. 483-524, 1976.
- [80] J. M. Kennedy and C. Ware, "Illusory contours can arise in dot figures," *Perception*, pp. 191-194, 1978.
- [81] G. T. Toussaint, "Pattern recognition and geometric complexity," in *Proc. 5th Int. Conf. Pattern Recognition* (Miami Beach), Dec. 1980, pp. 1324-1347.
- [82] V. Medek, "On the boundary of a finite set of points in the plane," *Computer Graphics and Image Processing*, vol. 15, pp. 93-99, 1981.
- [83] C. T. Zahn, "Graph-theoretical methods for detecting and describing gestalt clusters," *IEEE Trans. Comput.*, vol. 20, pp. 68-86, 1971.
- [84] G. Matheron, *Random Sets and Integral Geometry*. New York: Wiley, 1975.
- [85] J. Serra, *Image Analysis and Mathematical Morphology*. New York: Academic Press, 1982.
- [86] G. T. Toussaint, "Computational geometry and morphology," in *Proc. First Int. Symp. Science and Form* (Tsukuba, Japan), Nov. 1985, pp. 395-403.
- [87] J. Fairfield, "Contoured shape generation: Forms that people see in dot patterns," in *Proc. IEEE Conf. Cybernetics and Society*,

- 1979, pp. 60–64.
- [88] J. Fairfield, "Segmenting dot patterns by Voronoi diagram concavity," *IEEE Trans. Pattern Anal. Machine Intell.*, pp. 104–110, 1983.
 - [89] W. W. Moss, "Some new analytic and graphic approaches to numerical taxonomy, with an example from dermanyssidae (acari)," *Systematic Zoology*, vol. 16, pp. 177–207, 1967.
 - [90] H. Edelsbrunner and E. P. Mücke, "Three-dimensional alpha shapes," manuscript, 1992.
 - [91] B. Rosenberg and D. J. Langridge, "A computational view of perception," *Perception*, pp. 415–424, 1973.
 - [92] H. Edelsbrunner, G. Rote, and E. Welzl, "Testing the necklace condition for shortest tours and optimal factors in the plane," *Theoretical Computer Science*, vol. 66, pp. 157–180, 1989.
 - [93] E. R. Reinferberg, "A problem on circles," *Math. Gaz.*, vol. 32, pp. 290–292, 1948.
 - [94] P. Bateman and P. Erdős, "Geometrical extrema suggested by a lemma of Besicovitch," *Amer. Math. Monthly*, vol. 58, pp. 306–314, 1951.
 - [95] J. Bentley and T. Ottman, "Algorithms for reporting and counting geometric intersections," *IEEE Trans. Comput.*, vol. 28, pp. 643–647, 1979.
 - [96] G. Rote, "Two solvable cases of the traveling salesman problem," Ph.D. thesis, Technische Universität Graz, Austria, 1988.
 - [97] L. Guibas, J. Pach, and M. Sharir, "Generalized sphere-of-influence graphs in higher dimensions," manuscript, Tel-Aviv University, 1992.
 - [98] M. S. Jacobson, M. J. Lipman, and F. R. McMorris, "Trees that are sphere-of-influence graphs," Tech. Rep., University of Louisville, 1989.
 - [99] F. Harary, M. S. Jacobson, M. J. Lipman, and F. R. McMorris, "Sphere-of-influence graphs defined on prescribed graph," manuscript, University of Louisville, 1990.
 - [100] D. W. Dearholt, N. Gonzales, H. Kirpekar, "Associative network databases for computer vision," in *Proc. 21st Asilomar Conf. Signals, Systems, and Computers*, 1987, pp. 859–864.
 - [101] G. Kurup, "A database organized on the basis of similarities with applications in computer vision," Ph.D. thesis, New Mexico State University, June 1991.
 - [102] D. W. Dearholt, N. Gonzales, and G. Kurup, "Monotonic search networks for computer vision databases," in *Proc. 22nd Asilomar Conf. Signals, Systems, and Computers*, 1988, pp. 548–553.
 - [103] T. M. Cover and P. E. Hart, "Nearest neighbor pattern classification," *IEEE Trans. Inform. Theory*, vol. 13, pp. 21–27, 1967.
 - [104] L. P. Devroye, "On the inequality of Cover and Hart in nearest neighbor discrimination," *IEEE Trans. Pattern Anal. Machine Intell.*, vol. 3, pp. 75–78, 1981.
 - [105] P. M. Vaidya, "An $O(n \log n)$ algorithm for the all-nearest-neighbors problem," *Discrete and Computational Geometry*, pp. 101–115, 1989.
 - [106] J. H. Friedman, J. L. Bentley, and R. A. Finkel, "An algorithm for finding best matches in logarithmic expected time," *ACM Trans. Math. Software*, vol. 3, pp. 209–226, 1977.
 - [107] D. T. Lee and F. P. Preparata, "Location of a point in a planar subdivision and its applications," *SIAM J. Comput.*, pp. 594–606, 1977.
 - [108] D. T. Lee, "Rectangle proximity graphs and rectilinear shortest path problems," in *Proc. First Workshop Proximity Graphs. Memoranda in Computer and Cognitive Science MCCS-91-224*, Computing Research Laboratory, New Mexico State University, Las Cruces, 1991, pp. 57–87.
 - [109] L. P. Lefkovich, "Species associations and conditional clustering: clustering with or without pairwise resemblances, in *Developments in Numerical Ecology* (NATO ASI Series, vol. G14), P. Legendre and L. Legendre, Eds. New York: Springer-Verlag, 1987, pp. 309–331.
 - [110] L. P. Lefkovich, "Further nonparametric tests for comparing dissimilarity matrices based on the relative neighborhood graph," *Mathematical Biosciences*, vol. 73, pp. 71–88, 1985.
 - [111] L. P. Lefkovich, "A nonparametric method for comparing dissimilarity matrices, a general measure of biogeographical distance, and their applications," *Amer. Natur.*, vol. 123, pp. 484–499, 1984.
 - [112] P. F. Ash and E. D. Bolker, "Recognizing Dirichlet tessellations," *Geometriae Dedicata*, vol. 19, pp. 175–206, 1985.

- [113] M. Chang, C. Tang, and R. Lee, "Solving the Euclidean bottleneck biconnected edge subgraph problem by 2-relative neighborhood graph," *Discrete and Applied Mathematics*, vol. 39, pp. 1–12, 1992.
- [114] B. Sendov, "Planar neighborhood graphs without cycles," *Comptes rendus de l'Académie Bulgare des sciences*, vol. 44, no. 4, pp. 23–25, 1991.
- [115] F. Pernus, "The Delaunay triangulation and the shape hull as tools in muscle fiber analysis," *Pattern Recognition Lett.*, vol. 8, pp. 197–202, 1988.
- [116] T. Saito, J.-I. Toriwaki, and S. Yokoi, "Properties of extended digital α -hull with applications to shape feature analysis of a figure set," *Forma*, vol. 6, pp. 9–25, 1991.



Jerzy W. Jaromczyk received the Ph.D. in computer science in 1984 from Warsaw University, Poland.

Currently he is an associate professor in the Department of Computer Science at the University of Kentucky in Lexington. His research interests include algorithm design and applications.

Dr. Jaromczyk is a member of the Association for Computing Machinery and the European Association for Theoretical Computer Science.



Godfried T. Toussaint received the B.Sc. degree from the University of Tulsa, Tulsa, OK, and the M.A.Sc. and Ph.D. degrees from the University of British Columbia, Vancouver, B.C., Canada, in 1968, 1970, and 1972, respectively, all in electrical engineering.

Since 1972 he has been with the School of Computer Science at McGill University, teaching and doing research in the areas of information theory, pattern recognition, and computational geometry. During the summers of 1975 and 1977 he was a Visiting Scholar at the Information Systems Laboratory, Stanford University. The sabbatical year 1980–81 he spent as a Visiting Scientist at the Applied Mathematics Research Center of the University of Montreal. During the spring of 1986 he was a Visiting Scholar at the Courant Institute of Mathematical Sciences, at New York University. During the fall of 1988 he was a British Columbia Advanced Systems Institute Fellow at Simon Fraser University. In the spring of 1989 he was a Visiting Professor in the Mathematics Department at the University of West Indies, Cave Hill, Barbados, and during the summer of 1989 he was a Visiting Scientist in the Department of Mathematics and Computer Science at the University of Amsterdam.

Dr. Toussaint is a former council member of the North American Branch of the Classification Society and a former associate editor of the IEEE TRANSACTIONS ON INFORMATION THEORY and of the IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE. Presently he is an associate editor of the Plenum Press Series on Advanced Applications in Pattern Recognition and associate editor of the journals *Pattern Recognition*, *Computational Geometry: Theory and Applications*, *The International Journal of Computational Geometry and Applications*, and *The Visual Computer*. He is also on the editorial boards of the journals *Discrete & Computational Geometry* and *Forma* and on the advisory board of the IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE. He is a member of several learned societies, including the Pattern Recognition Society and the New York Academy of Sciences. He is the editor of two books published by North Holland: *Computational Geometry* (1985) and *Computational Morphology* (1988), and he served as guest editor of a special issue of *The Visual Computer* on computational geometry (May 1988). In 1978 he was the recipient of the Pattern Recognition Society's Best Paper of the Year Award and in 1985 he was awarded a Killam Fellowship by the Canada Council to carry out a two-year research project on the movable separability of sets.